

My Name

linkedin.com/in/my-name | github.com/MyName

Education

T30 University – Bachelor of Science in Computer Science
GPA: 3.85/4.0

Expected April 2028

Projects

Custom 64-bit x86-64 Operating System | [GitHub](#)

- Collaborated with 4 developers to build a 64-bit operating system using C and the Limine bootloader, implementing CPU structures (GDT, TSS).
- Designed framebuffer graphics, implementing bitmap font rendering and automatic terminal scrolling.
- Created formatted output library with custom vsnprintf, enabling debugging through serial COM1 interface.

Custom 32-bit x86 Bootloader with EHCI Driver | [GitHub](#)

- Developed two-stage x86 bootloader that transitions CPU from real mode to protected mode, performing memory relocation and loading 4KB kernel from disk via BIOS interrupts.
- Built USB 2.0 controller driver in C that enumerates connected devices through PCI bus scanning and USB control transfers, managing device communication via MMIO registers.
- Implemented VGA text rendering and serial debugging interface with millisecond-precision timing to provide real-time feedback during hardware initialization.

Custom 64-bit x86 Bootloader with AHCI (SATA) Driver | [GitHub](#)

- Built 64-bit x86 bootloader that configures page tables for 1GB memory addressing and transitions CPU through real, protected, and long modes to launch the kernel.
- Developed AHCI driver in C that detects SATA storage devices and executes disk read/write operations using DMA transfers, with BIOS handoff for OS-level control.
- Developed MMIO interface for hardware communication, including device enumeration and metadata extraction (model, serial, LBA48 capacity) with VGA/serial output.

Experience

Software Developer, Hair & Grooming Studio – City, State

July 2023 – June 2025

- Developed appointment booking system using Node.js and PostgreSQL with Redis caching layer that reduced database query overhead by 40%.
- Implemented form validation using vanilla JavaScript to prevent malformed requests, reducing invalid submissions by 30% and lowering unnecessary backend processing.
- Built a lightweight monitoring dashboard using Chart.js to track appointment throughput and system usage trends, supporting scheduling efficiency and overbooking incidents.

Skills

Languages: C, x86/x86-64 Assembly, Python, JavaScript

Systems Programming: Bootloader development, OS kernel design, bare-metal programming, interrupt handling

Hardware Protocols: AHCI (SATA), EHCI (USB 2.0), UART, PCI/PCIe

Tools & Debuggers: NASM, QEMU, GDB, Git